

RANDOM TRANSLATION INTERSECTIONS ON BINARY VECTOR SPACES: SHARP SPAN CLOCKS AND STABILIZER-WEIGHTED FIBRES

ANONYMOUS

ABSTRACT. Let $V = \mathbb{F}_2^d$ and repeatedly replace a subset $A \subseteq V$ by $A \cap (A + v)$, where v is sampled uniformly from V . A length- t history acts exactly as intersection by every translate in the span of its sampled vectors. We use this sufficient statistic to give the complete rank law and a sharp worst-non-full-source emptying clock, witnessed by $V \setminus \{0\}$. For every target B , every time, and every source size, we then enumerate source-history pairs by a polynomial determined by the translation stabilizer of B . Its one-step boundary separates targets with trivial stabilizer and recovers the stabilizer dimension once the phase dimension and target size are known. The forward erosion algebra and finite-field rank law are classical inputs; the contribution assessed here is the exact conjunction with the target-dependent inverse atlas. All release claims remain under `HOLD_EXTERNAL`.

1. PROCESS AND THEOREM

Write $A + v = \{a + v : a \in A\}$. Starting from $A_0 \in \mathcal{P}(V)$, sample independent uniform vectors $v_1, v_2, \dots$ and set

$$(1) \quad A_i = A_{i-1} \cap (A_{i-1} + v_i).$$

For a subspace $H \leq V$, define

$$E_H(A) = \bigcap_{h \in H} (A + h), \quad H_t = \langle v_1, \dots, v_t \rangle.$$

Let

$$(2) \quad S(t, r) = \prod_{i=0}^{r-1} (2^t - 2^i),$$

with the empty product equal to one, and put $S(t, r) = 0$ for $r > t$. We write $\begin{bmatrix} m \\ r \end{bmatrix}_2$ for the Gaussian binomial coefficient.

For a target $B \subseteq V$, set $b = |B|$ and

$$\text{Stab}(B) = \{v \in V : B + v = B\}, \quad s = \dim \text{Stab}(B).$$

The joint source-size/history polynomial is

$$(3) \quad F_t(B; z) = \sum_{A \subseteq V} \sum_{(v_1, \dots, v_t) \in V^t} \mathbf{1}\{A_t = B\} z^{|A|}.$$

Theorem 1 (Span clock and every-target inverse atlas). *For every $d, t \geq 0$ the following statements hold.*

(1) *Every history satisfies the pointwise identity*

$$(4) \quad A_t = E_{H_t}(A_0).$$

Moreover,

$$(5) \quad \Pr(\dim H_t = r) = \begin{bmatrix} d \\ r \end{bmatrix}_2 \frac{S(t, r)}{2^{dt}}.$$

(2) *The only states fixed by every possible sampled update are $\emptyset$ and V . Let $\sigma = \min\{t : H_t = V\}$. Every $A_0 \neq V$ is empty by time σ , and this bound is attained by $A_\star = V \setminus \{0\}$. Thus the sharp worst-source emptying distribution is*

$$(6) \quad \Pr(\sigma \leq t) = \begin{cases} 0, & t < d, \\ \prod_{i=0}^{d-1} (1 - 2^{i-t}), & t \geq d, \end{cases}$$

and

$$(7) \quad \mathbb{E}\sigma = \sum_{r=0}^{d-1} \frac{1}{1 - 2^{r-d}}.$$

(3) For every target B and indeterminate z ,

$$(8) \quad F_t(B; z) = z^b \sum_{r=0}^s \begin{bmatrix} s \\ r \end{bmatrix}_2 S(t, r) \left((1+z)^{2^r} - z^{2^r} \right)^{2^{d-r}-b/2^r}.$$

For $r \leq s$, the exponent is a nonnegative integer. Formula (8) includes $t = 0$, $d = 0$, and the empty and full targets. At one step its unweighted specialization is the boundary-safe formula

$$(9) \quad F_1(B; 1) = \begin{cases} 1, & s = 0, \\ 1 + (2^s - 1)3^{2^{d-1}-b/2}, & s \geq 1. \end{cases}$$

(4) The phase cardinality $|\mathcal{P}(V)| = 2^{2^d}$ determines d . At fixed (d, b) , the value in (9) is strictly increasing over the feasible values of s , with value one occurring exactly at $s = 0$. Consequently phase size, target size, and one-step inverse mass recover the target stabilizer dimension.

The clock in part (2) is a sharp universal bound, not an assertion that every non-full source empties exactly at σ . This distinction is important: individual sources may disappear earlier.

2. HISTORY SPANS AND THE SHARP CLOCK

The algebra of erosions by translates is standard in mathematical morphology [2, 3]; stochastic morphological sampling is also established background [4]. We record the short reduction needed for this particular process.

Lemma 2. For subspaces $H, K \leq V$, $\mathbf{E}_K(\mathbf{E}_H(A)) = \mathbf{E}_{H+K}(A)$.

Proof. Distributing intersections over translation gives

$$\mathbf{E}_K(\mathbf{E}_H(A)) = \bigcap_{k \in K} \bigcap_{h \in H} (A + h + k) = \bigcap_{u \in H+K} (A + u).$$

Repeated factors do not affect an intersection. $\square$

Proof of Theorem 1(1). Because $\langle v \rangle = \{0, v\}$ in characteristic two, one update is $\mathbf{E}_{\langle v \rangle}$. Lemma 2 iterated over the history proves (4).

For a fixed r -subspace H , histories spanning H are surjective linear maps $\mathbb{F}_2^t \rightarrow H$. Choosing the images dually as an ordered independent r -tuple in $\mathbb{F}_2^t$ gives (2). There are $\begin{bmatrix} d \\ r \end{bmatrix}_2$ choices of H , which proves (5). This finite-field rank enumeration is classical; we do not claim it as a standalone increment [1]. $\square$

Proof of Theorem 1(2). If A is fixed by every update, then $A \subseteq A + v$ for every v . Cardinalities force equality, so A is invariant under all translations and hence is either $\emptyset$ or V .

When $H_t = V$, the set $\mathbf{E}_V(A)$ is V for $A = V$ and is empty otherwise. Sharpness follows from the stronger identity

$$(10) \quad \mathbf{E}_H(V \setminus \{0\}) = V \setminus H :$$

x survives exactly when $x + h \neq 0$ for every $h \in H$, equivalently when $x \notin H$. Therefore the witness empties exactly at full span.

Equation (6) is the full-rank probability in (5). At rank $r < d$, a fresh uniform vector increases the rank with probability $1 - 2^{r-d}$. The independent geometric waiting times for the successive rank increments have means $(1 - 2^{r-d})^{-1}$, proving (7). $\square$

3. STABILIZER-WEIGHTED INVERSE FIBRES

The inverse enumeration uses information absent from the rank marginal: the specific target controls which history spans are possible.

Lemma 3 (Fixed-span fibre). *Let $H \leq V$ have dimension r . There exists a source A with $E_H(A) = B$ if and only if $H \leq \text{Stab}(B)$. In that case the source-size polynomial is*

$$(11) \quad \sum_{A: E_H(A)=B} z^{|A|} = z^b \left((1+z)^{2^r} - z^{2^r} \right)^{2^{d-r} - b/2^r}.$$

Proof. The set $E_H(A)$ is the union of precisely those H -cosets wholly contained in A . Hence it is H -invariant, proving necessity. Conversely, if $H \leq \text{Stab}(B)$, then B is a union of $b/2^r$ full H -cosets. A source with erosion B must contain all of them. On each of the remaining $2^{d-r} - b/2^r$ cosets it may choose any proper subset, independently. One coset contributes $(1+z)^{2^r} - z^{2^r}$, which proves (11). $\square$

Proof of Theorem 1(3). For a rank- r history, Lemma 3 permits exactly the subspaces $H \leq \text{Stab}(B)$. Their number is $\begin{bmatrix} s \\ r \end{bmatrix}_2$, and each is generated by $S(t, r)$ ordered histories. Summing (11) proves (8).

When $t = 0$, only $r = 0$ remains and the expression reduces to z^b , as it must. If $B = V$, the only source is V and the rank sum gives $F_t(V; z) = 2^{dt} z^{2^d}$. The empty target and $d = 0$ require no exceptional convention beyond empty products.

At $t = 1$, the zero history has rank zero and contributes one at $z = 1$. Every nonzero vector of $\text{Stab}(B)$ spans a rank-one subspace, and each outside two-point coset has three proper subsets. There are $2^s - 1$ such histories. If $s \geq 1$, then B is a union of two-point orbits, so b is even and the second line of (9) follows. If $s = 0$, there is no nonzero stabilizing history and the value is one. Keeping this branch separate is essential for odd b , where the other displayed exponent would not be an integer. $\square$

Proof of Theorem 1(4). The map $d \mapsto 2^{2^d}$ is injective. At fixed (d, b) the power of three in the second line of (9) is constant, while $2^s - 1$ is strictly increasing. Its positive contribution also separates every $s \geq 1$ from the value one at $s = 0$. $\square$

The polynomial contains more than unweighted fibre sizes: its coefficient of z^k counts all length- t histories and k -point sources reaching the prescribed target. Thus target shape enters through $\text{Stab}(B)$ even when target cardinality is held fixed.

4. SOURCE SUBTRACTION, CONTROLS, AND SCOPE

The update's erosion interpretation and composition algebra receive zero contribution credit [2, 3]; so do the generic stochastic-morphology setting [4] and the finite-field rank distribution [1]. The bounded source audit found no primary source for the exact conjunction of the sharp witness (10), the arbitrary-target polynomial (8), and the recovery law (9). This non-hit is not a novelty, priority, or freedom-to-publish claim. Internally, the closest comparators use deterministic linear-subspace descent or random graph-cut signatures; neither transfers the affine-coset proper-subset fibre proof.

A paper-local, standard-library verifier applies every literal update and independently evaluates the span erosion over finite exhaustive boxes. It compares every target and source-size coefficient with (8), checks the rank census and full-rank boundary, tests the witness over all subspaces through dimension six, and exercises $d = 0$, $t = 0$, empty/full targets, odd-cardinality trivial stabilizers, and the repaired one-step branch. The frozen assertion total and transcript digest are recorded in the accompanying canonical file and build record. The run executes 1,712,974 integer assertions; two fresh replays were byte-identical. The canonical transcript has SHA-256 (the following two lines concatenate):

```
c31ec0a098bab52241eb2765bd6fef06
69fdacdb4486ca69bea9dfc56fbab62b.
```

Enumeration supplies counterexample pressure only; the all-parameter conclusions rest on the proofs above.

LIMITATIONS

The process uses independent uniform translations of a binary vector space and begins from an arbitrary labelled subset. We do not treat biased or dependent histories, non-elementary finite groups, noisy observations, unlabelled targets, or asymptotic distributional limits. The source search was bounded, so a direct owner under different terminology would reopen the claim boundary.

DATA AVAILABILITY

No external data were used. The deterministic verifier and its canonical transcript provide the complete paper-local exact control.

ETHICS STATEMENT

This mathematical study involved no human participants, animals, personal data, or field intervention.

AUTHOR CONTRIBUTIONS

The anonymous author performed the derivation, proof, exact checks, source-boundary audit, and manuscript preparation.

CONFLICT OF INTEREST

The author declares no conflict of interest.

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EXTERNAL STATUS

This anonymous internal artifact remains `HOLD_EXTERNAL`. It is not cleared for posting, submission, circulation, or author contact.

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